

24	A	Negative charges gain KE as it loses EPE from Source S to the hollow metal container. $qV = \frac{1}{2} mv^2 \Rightarrow 2V (q/m) = v^2 \text{ --- (1)}$ Inside the hollow metal container, in order for the ions to pass through un-deviated, $F_B = F_E$ $Bqv = qE = q(V/d)$ $Bv = (V/d)$ $v = (V/Bd) \text{ --- (2)}$ Sub (2) in (1) $(V/Bd)^2 = 2V (q/m)$ $q/m = V / (2 B^2 d^2)$

25	C	Emf of a rotating disc of radius l